

Technical Datasheet Input / Output Modules with MQTT Protocol with Ethernet Interface

The **IO Module** communicates via an **Ethernet Interface** and is designed to interface seamlessly with other **MQTT-compatible devices**. The module acts as an **input/output device** that bridges the **MQTT network** and the connected gateway, enabling real-time data exchange over the MQTT protocol. This **Digital I/O module** is robust, energy-efficient, and user-friendly, making it ideal for a wide range of IoT applications. It supports for **digital inputs**, and allows users to easily publish and subscribe to topics on an MQTT broker, facilitating communication with other devices, cloud platforms, and remote monitoring systems.

16 Port DI Module with Modbus TCP Ethernet: -



The IO modules enclosures are DIN rail Mountable. It's a closed Enclosure with openings for connectors and LED indicators.

The design of the modules incorporates '**resettable Fuses**' to safeguard against reverse polarity connection for **Power Input**.

There are LED indicators for each of the IO ports as well as for the module Power.

Specifications

General –

I/O Connectors	22 Pin 5.08 mm pitch pluggable screw Terminal Block, 2 Pin 5.08 mm pitch pluggable screw terminals.
Dimensions	110 mm L x 110 mm B x 50 mm H
Power	Input Power – 12-24 V DC Typical – 12V DC @ 120mA
Operating Temperature	0 – 60° C (32 ~ 140°F)
Storage Temperature	-20 - 70° C (-4 ~ 158°F)
Storage Humidity	5 ~ 95 % RH, non – Condensing

Certifications



DI Inputs –

Channels	16
Sense Voltage	3.3 – 30 VDC
Sense Logic	Logic '0' = <1 VDC, Logic '1' = >3.3 VDC
Isolation	Optically isolated
Response Time	2 msec, Max 6 msec

Technical Specifications:

Key Features

- **MQTT Protocol Support:** Fully compliant with the MQTT protocol, supporting both MQTT 3.1.1 and MQTT 5.0.
- **Ethernet Connectivity:** Standard Ethernet port (10/100 Mbps) for stable and high-speed data transmission.
- **Low Power Consumption:** Optimized for low power usage, ideal for battery-operated or energy-efficient applications.
- **Easy Integration:** Supports standard libraries and tools for easy integration with existing systems.
- **Message Retention & Quality of Service (QoS):** Supports various QoS levels (0, 1, 2) to guarantee message delivery.
- **Secure Communication:** Support for TLS/SSL encryption for secure communication with MQTT brokers.

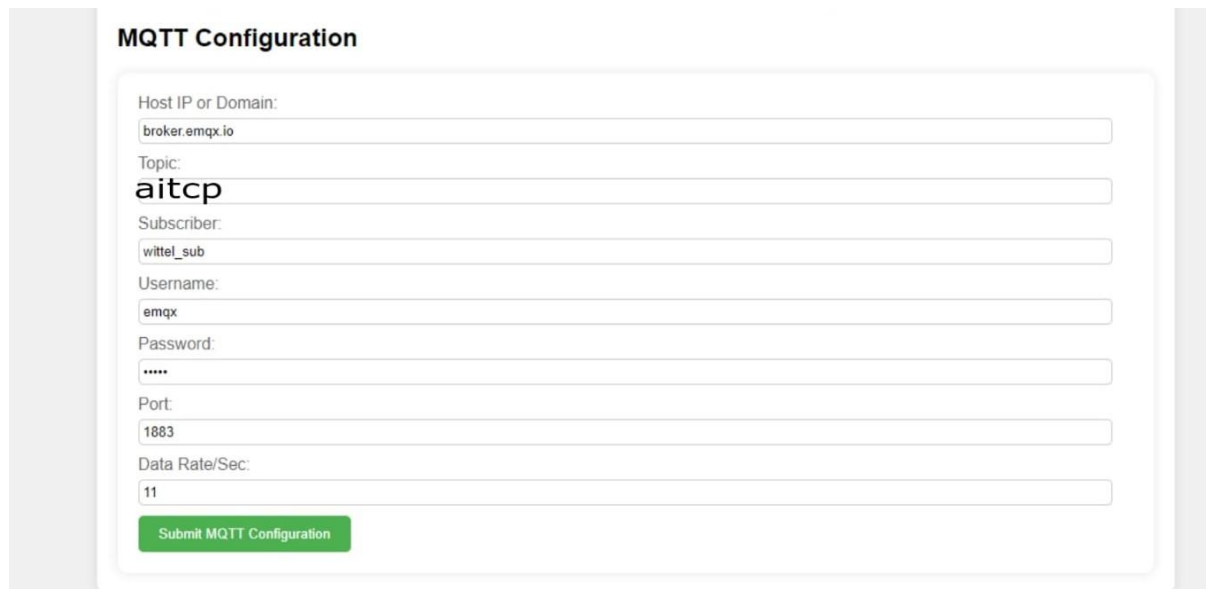
- **Real-time Data Handling:** Efficient, real-time messaging with minimal delay.
- **Compact Form Factor:** Small and lightweight, designed for embedded systems.

- **Configuration Process:-**

- For Configuration Enter IP 192.168.1.40.
- You will get this type of page as shown in the image(1.1)

- After Entering the ip You will see interface as shown in the image.

Note:- (For more help you can refer below image1.1 for login)



The image shows a web interface titled "MQTT Configuration". It contains several input fields for configuring an MQTT client. The fields are labeled as follows: "Host IP or Domain:" with the value "broker.emqx.io"; "Topic:" with the value "aitcp"; "Subscriber:" with the value "wittel_sub"; "Username:" with the value "emqx"; "Password:" with masked characters "*****"; "Port:" with the value "1883"; and "Data Rate/Sec:" with the value "11". At the bottom of the form is a green button labeled "Submit MQTT Configuration".

Login and MQTT Configuration : -

1. Login Prompt

When you open the device's details page, you will be prompted to enter a **username** and **password**.

2. Default Username

The default **username** is usually admin.

3. Default Password

The default **password** is typically admin or password.

4. MQTT Port

The default MQTT port is **1883** for unencrypted connections.

For secure MQTT over TLS, use **port 8883**.

5. Topic Name

Ensure you configure the correct **topic name** where the analog and digital data will be published.

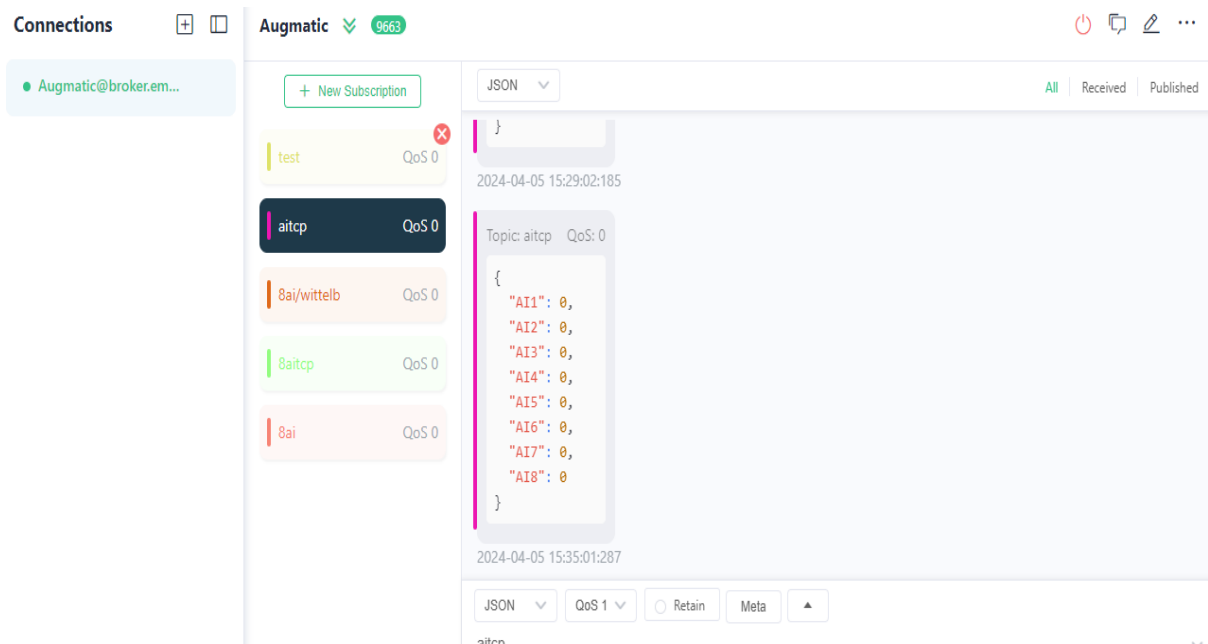
For example, if you set the topic to “aitcp”, you will receive analog and digital data on that topic.

6. Data Rate

The **data rate** refers to how often the device publishes messages. You can set it to a specific interval, such as every second or every few seconds, depending on how often you want the data to update.

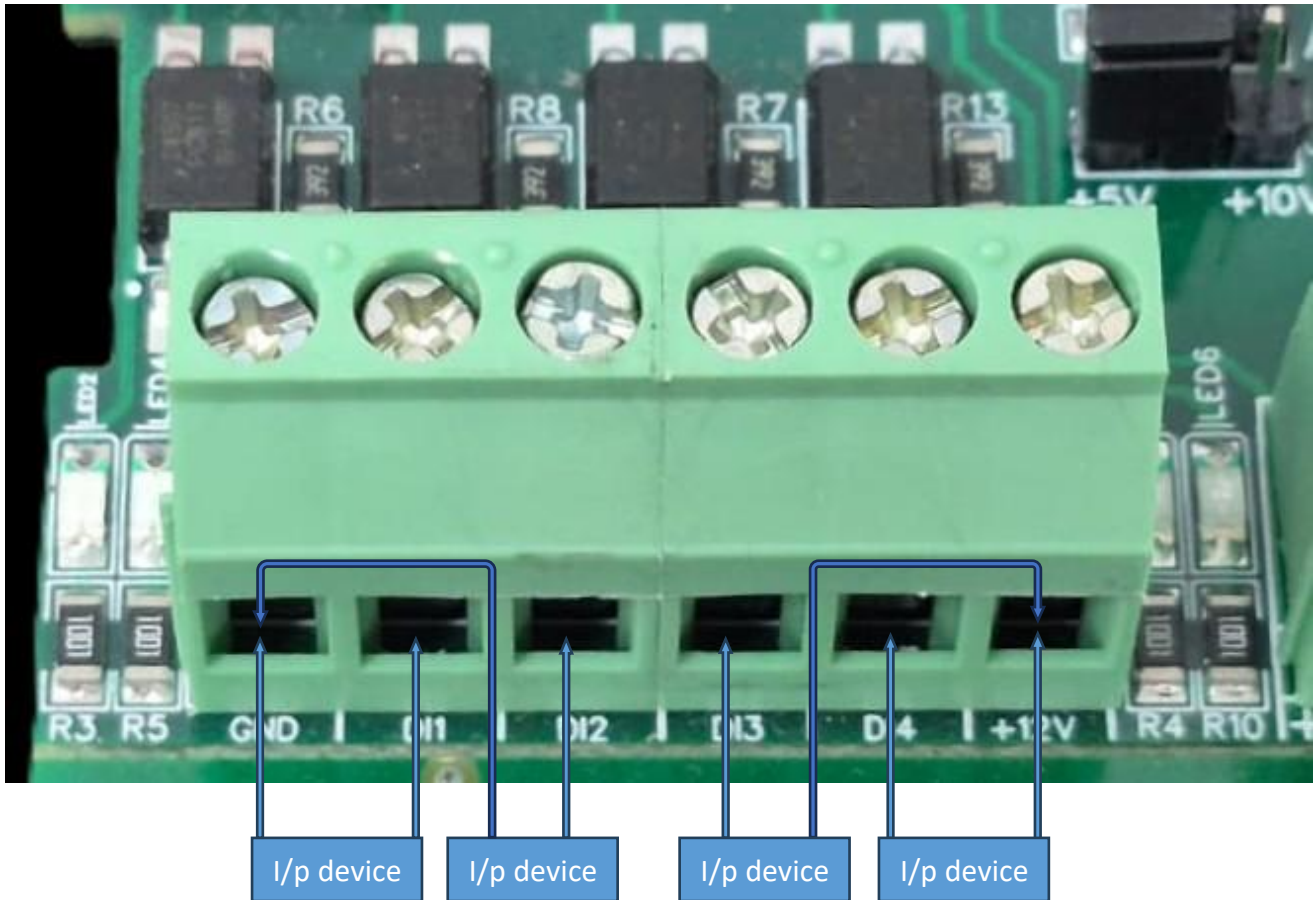
Note: (Username Password is optional).

- After submitting the data you will get data on on mqtt cloud.

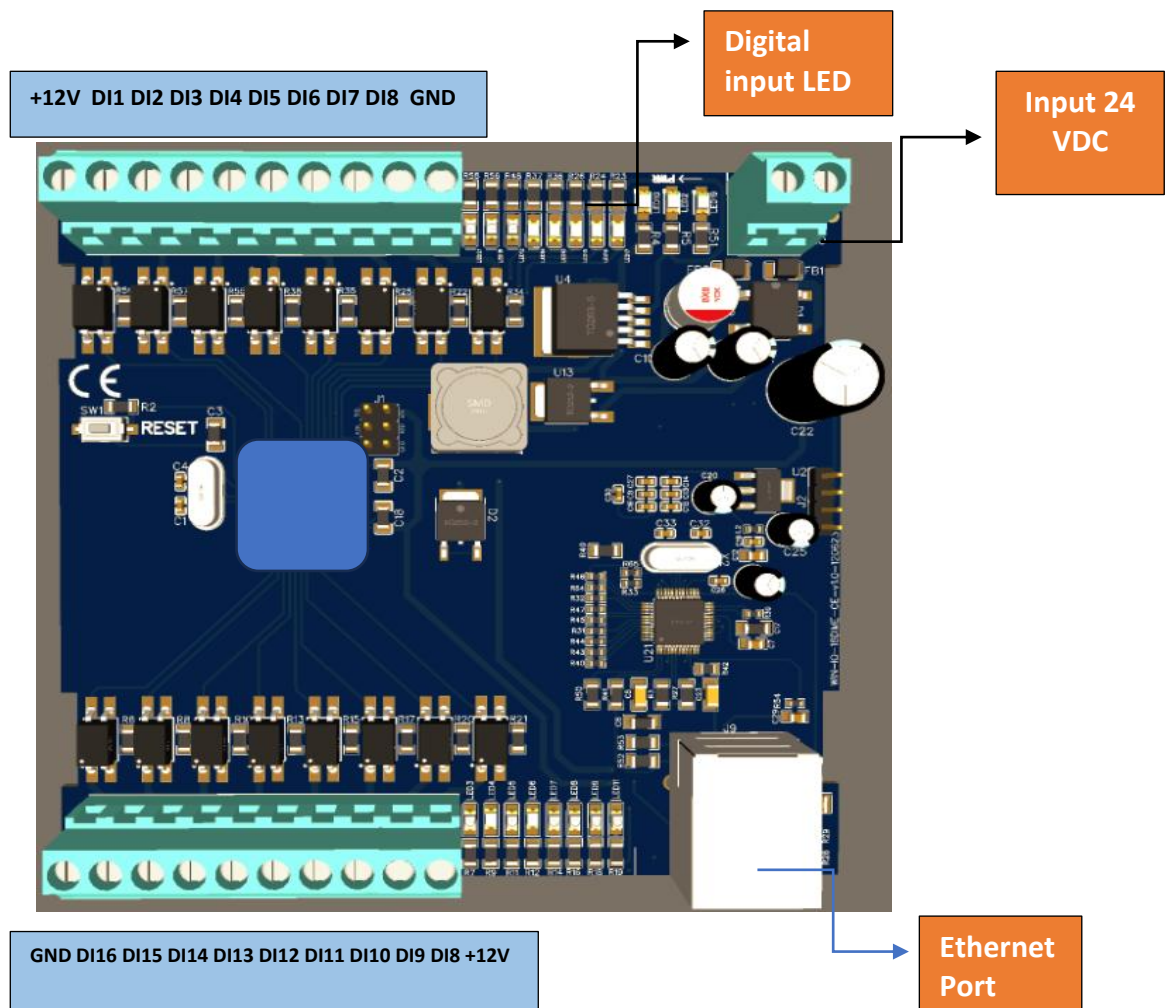


- Here you can see the data of analog input data same as you can see digital input data of 16 channel..

DI



- **Operation:** The module provides four isolated input channels (**DI1–DI4**) for sensing external signals or dry contacts. These inputs are opto-isolated for circuit protection.
- **Terminal Layout:**
 - **GND:** Common Ground reference.
 - **DI1 – DI4:** Individual Digital Input channels.
 - **+12V:** Internal reference voltage for sourcing "Dry Contact" switches.
- **Wiring Options:**
 - **Dry Contact:** Connect a switch between the **+12V** terminal and any **DI** pin.
 - **External Voltage:** Connect the external signal to a **DI** pin and the external ground to the **GND** terminal.
- **Voltage Rating:** The input circuitry is rated for a maximum of **80VDC**. Ensure external signals do not exceed this limit to prevent damage to the isolation ICs.



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